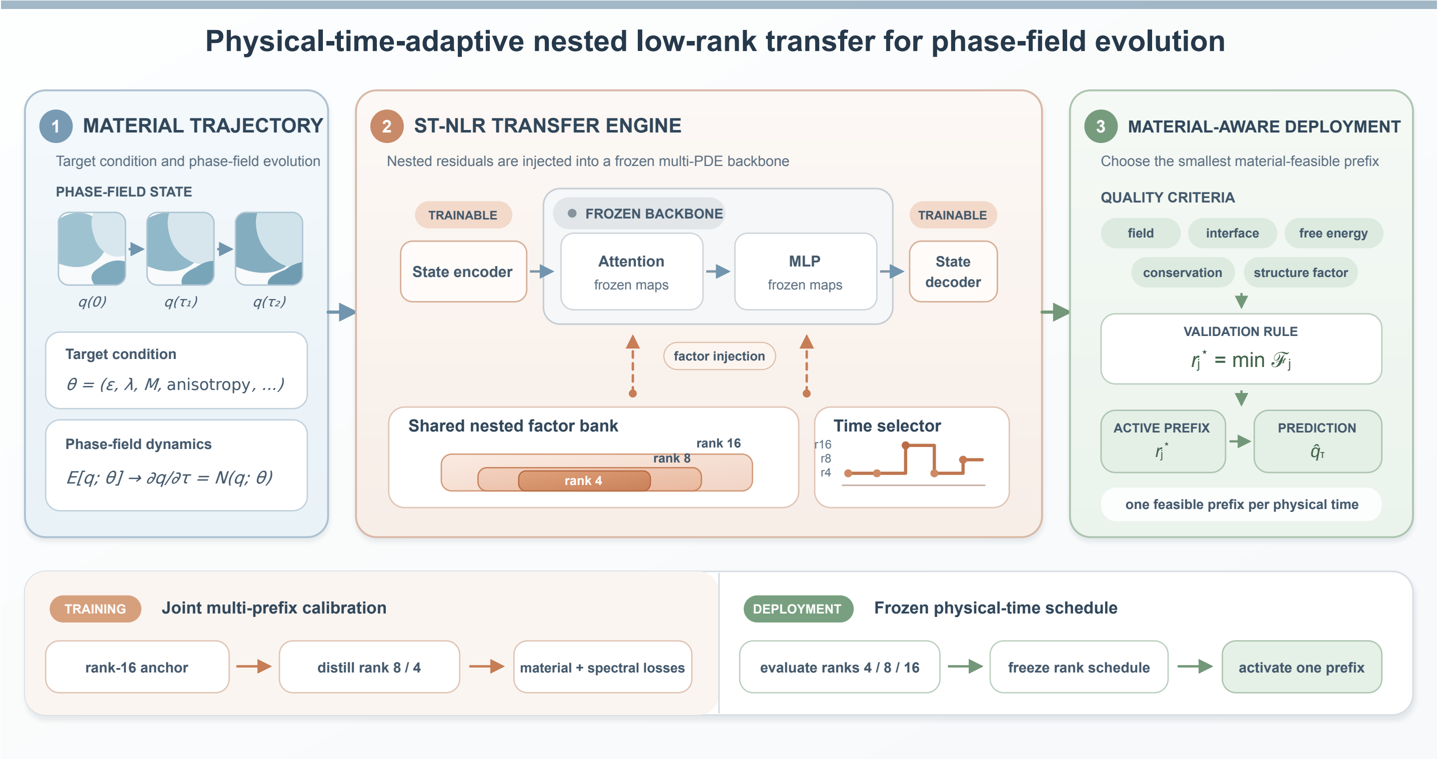

Physical-time-adaptive nested low-rank transfer for phase-field evolution
1 MATERIAL TRAJECTORY
Target condition and phase-field evolution
PHASE-FIELD STATE
q(0)
q(τ₁)
q(τ₂)
Target condition
θ = (ε, λ, M, anisotropy, ...)
Phase-field dynamics
E[q; θ] → ∂q/∂τ = N(q; θ)
2 ST-NLR TRANSFER ENGINE
Nested residuals are injected into a frozen multi-PDE backbone
TRAINABLE
State encoder
FROZEN BACKBONE
Attention
frozen maps
MLP
frozen maps
TRAINABLE
State decoder
factor injection
Shared nested factor bank
rank 16
rank 8
rank 4
Time selector
r16
r8
r4
3 MATERIAL-AWARE DEPLOYMENT
Choose the smallest material-feasible prefix
QUALITY CRITERIA
field
interface
free energy
conservation
structure factor
VALIDATION RULE
rj* = min ℱj
ACTIVE PREFIX
rj*
PREDICTION
q̂τ
one feasible prefix per physical time
TRAINING
Joint multi-prefix calibration
rank-16 anchor
distill rank 8 / 4
material + spectral losses
DEPLOYMENT
Frozen physical-time schedule
evaluate ranks 4 / 8 / 16
freeze rank schedule
activate one prefix